

CSDS 312 Project: Final Report

**Demographic and Socioeconomic Factors in Social Media
Addiction: Analyzing Behavioral Patterns and Influences**

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I. Abstract

This study examines how social media addiction varies across different demographics and income levels. By analyzing behavioral patterns, psychological factors, and platform engagement strategies, we explore how age, gender influence susceptibility to social media addiction. Using different machine learning models, including Random Forest Classifier, XGBoost, and Logistic Regression, to predict the addictiveness of social media in different demographics. The research highlights how algorithm-driven content, social validation, and economic disparities shape user behavior, leading to varying degrees of dependence. Understanding these differences can help develop targeted interventions to promote healthier social media usage and digital well-being across diverse populations.

II. Introduction

A. Background and Context

In the digital age, social media has become deeply embedded in daily routines, transforming how individuals interact, share information, and shape their identities. Platforms like Instagram, TikTok, and Facebook influence public discourse, cultural trends, and emotional well-being. While these platforms provide benefits such as connectivity and creativity, excessive usage has raised concerns about addiction, particularly among younger users. Social media addiction, characterized by compulsive usage and psychological dependency, is now recognized as a growing behavioral issue.

B. Research Problem

The global prevalence of social media addiction varies significantly across cultural contexts. A meta-analysis encompassing over 34,000 participants from 32 countries revealed that collectivist societies exhibited higher addiction rates (31%) compared to individualist societies (14%), suggesting that cultural values and societal norms play a significant role in shaping online behavior. (Chen et al., 2021). IN addition to cultural differences, socioeconomic disparities have emerged as a critical factor influencing social media usage patterns. Research has shown that individuals from lower socioeconomic status (SES) backgrounds are more susceptible to addictive behaviors, with psychological

mediators such as loneliness, impulsivity, and feelings of alienation contributing to this vulnerability (Yang et al., 2024; Zhang et al., 2020)

C. Research Question

Given these insights, this study seeks to explore how demographic variables - including age, gender - interact with psychological and behavioral patterns to influence susceptibility to social media addiction. By applying machine learning models such as Logistic Regression, Random Forest Classifier, and XGBoost, we aim to identify the most predictive features associated with addictive behaviors and evaluate how these patterns vary across different population segments. Ultimately, this research seeks to contribute to a more nuanced understanding of the digital divide, offering data-driven insights that can inform targeted interventions and promote healthier digital habits.

This project seeks to answer: ***Can behavioral and demographic factors accurately predict a user's risk of social media addiction?***

D. Hypothesis

We hypothesize that users with higher engagement metrics (screen time, likes, posts) and certain demographic profiles (eg, young age, female gender) are more likely to exhibit signs of addiction. We employ data science methods to uncover the most predictive features and propose data-driven interventions. Understanding the mechanisms and risk factors behind social media addiction enables the creation of more effective interventions. By using data science techniques, this study not only identifies key predictors but also empowers stakeholders, including educators, mental health professionals, and platform developers, with the knowledge to promote healthier online habits.

III. Development Tools

A. Python

Python is a high-level general-purpose programming language that was conceived in the late 1980s. As one of the most popular programming languages used for a wide range of applications, including web development, data analysis, automation, and more. Due to its extensive libraries and frameworks, Python is widely used in the fields of AI and machine learning. Our group chose to work on most of the coding portion in this programming space, as it has extensive libraries and frameworks that allow us to implement different types of machine learning techniques to allow comparison with the results of each model.

B. Pandas

Pandas is a software library for the Python programming language that is used for the manipulation and analysis of the data. It provides data structures which are efficient for working with structured data, such as tables and time series.

C. Scikit-learn

Sklearn is a software machine learning library for the Python programming language. The library is used for various machine learning and statistical model tasking. Sklearn supports different classification, regression, and clustering algorithms.

D. XGBoost

XGBoost, also known as extreme gradient boosting, is a powerful and widely used open-source machine learning library that implements the gradient boosting algorithm. XGBoost works by sequentially building decision trees, with each new tree attempting to correct the errors made by the previous ones, while the gradient boosting algorithm uses gradient descent to minimize a loss function and incorporate regularization techniques to prevent over-fitting.

E. Random Forest Classifier

Random Forest Classifier is a machine learning algorithm that uses an ensemble of decision trees to make predictions. It combines the predictions of multiple trees, trained on different random subsets of the data, to improve accuracy and reduce overfitting. The final classification is determined by the class with the most votes from the individual trees.

F. Logistic Regression

Logistic regression is a supervised machine learning algorithm used for binary classification tasks. It creates a model that estimates the relationship between binary independent variables and the probability of the event occurring.

G. Kaggle

To conduct our analysis, we sourced data from Kaggle, a Google-owned online platform widely used by data scientists and machine learning practitioners for accessing datasets, participating in competitions, and sharing code. Kaggle hosts a large collection of publicly available datasets across various domains, including health, technology, finance, and social behavior.

IV. Methodology

A. Dataset Sources

The data cleaning and preprocessing phase was crucial for ensuring the quality and reliability of our analysis. The process involved several key steps:

1. *Data Loading and Initial Assessment*

We utilized two public datasets sourced from Kaggle:

- Social Media Usage and Emotional Well-Being: This dataset contains individual-level data on daily platform usage, emotional self-assessment, and engagement metrics (likes, posts, follows, etc.).
- Social Media Usage Dataset (Applications): This dataset includes demographic information such as age, gender, along with behavioral data on platform activity.

The datasets were loaded into Pandas DataFrames. We performed an initial scan to evaluate dataset shape, check for missing values, and inspect column data types. This step helped us identify inconsistencies, null entries, and formatting issues:

```
# Load the dataset
df = pd.read_csv('social_media_usage.csv')

# Initial data assessment
print(f"Dataset shape: {df.shape}")
print("\nMissing values per column:")
print(df.isnull().sum())
print("\nData types:")
print(df.dtypes)
```

Fig. 1: Data loading and initial assessment

B. Data Cleaning and Feature Engineering

Our preprocessing pipeline followed a structured sequence of operations:

1. *Handling Missing Values*

- Numerical columns were imputed using the median to preserve the original distribution without being skewed by outliers.
- Categorical columns were filled using the mode to retain the most representative values.

This approach maintained the statistical integrity of the dataset while preventing information loss.

```
def handle_missing_values(df):  
    # Fill missing numeric values with median  
    numeric_cols = df.select_dtypes(include=['int64', 'float64']).columns  
    for col in numeric_cols:  
        df[col] = df[col].fillna(df[col].median())  
  
    # Fill missing categorical values with mode  
    categorical_cols = df.select_dtypes(include=['object']).columns  
    for col in categorical_cols:  
        df[col] = df[col].fillna(df[col].mode()[0])  
  
    return df
```

Fig. 2: Handling missing values

2. *Data Type Conversion and Standardization*

- Categorical columns such as Platform and Gender were explicitly cast to categorical types.
- The Age column was converted to numeric, coercing non-numeric values to NaN, which were then handled as missing values.
- Numeric features were standardized using StandardScaler from Scikit-learn, ensuring that all inputs were on a common scale for training machine learning models.


```
def standardize_data(df):
    # Convert categorical variables to appropriate types
    df['Platform'] = df['Platform'].astype('category')
    df['Gender'] = df['Gender'].astype('category')

    # Convert Age to numeric, handling any non-numeric values
    df['Age'] = pd.to_numeric(df['Age'], errors='coerce')

    # Standardize numeric columns
    numeric_cols = df.select_dtypes(include=['int64', 'float64']).columns
    scaler = StandardScaler()
    df[numeric_cols] = scaler.fit_transform(df[numeric_cols])

    return df
```

Fig. 3: Data Type Conversion and Standardization

3. Feature Engineering

We introduced several engineered features to enhance predictive capacity:

- Usage_Hours was derived by converting daily usage minutes into hours.
- Usage_Category grouped users into "Low," "Moderate," "High," and "Very High" categories based on time spent.
- Age_Usage_Interaction was calculated as a product of age and usage time to capture complex behavioral dynamics across age groups.

```
def create_features(df):
    # Create time-based features
    df['Usage_Hours'] = df['Daily_Usage_Time (minutes)'] / 60
    df['Usage_Category'] = pd.cut(
        df['Usage_Hours'],
        bins=[0, 2, 4, 6, float('inf')],
        labels=['Low', 'Moderate', 'High', 'Very High']
    )

    # Create interaction features
    df['Age_Usage_Interaction'] = df['Age'] * df['Usage_Hours']

    return df
```

Fig. 4: Feature Engineering

4. Outlier Detection and Treatment

- We applied the Interquartile Range (IQR) method to identify outliers in numerical features.

- Instead of removing outliers, we capped them at the upper and lower bounds to preserve the dataset size while reducing skew.

```
def handle_outliers(df):
    numeric_cols = df.select_dtypes(include=['int64', 'float64']).columns

    for col in numeric_cols:
        # Calculate IQR
        Q1 = df[col].quantile(0.25)
        Q3 = df[col].quantile(0.75)
        IQR = Q3 - Q1

        # Define bounds
        lower_bound = Q1 - 1.5 * IQR
        upper_bound = Q3 + 1.5 * IQR

        # Cap outliers
        df[col] = df[col].clip(lower_bound, upper_bound)

    return df
```

Fig. 5: Outlier Detection and Treatment

5. Final Data Preparation

All the above preprocessing steps were encapsulated in a final preparation function. This function processed the raw data through missing value treatment, standardization, feature engineering, and outlier handling. A new target variable `addiction_level` was created using quantile-based binning of daily usage time into four risk categories: Low, Moderate, High, and Severe Risk.

```
def prepare_data(df):
    # Apply all preprocessing steps
    df = handle_missing_values(df)
    df = standardize_data(df)
    df = create_features(df)
    df = handle_outliers(df)

    # Create target variable
    df['addiction_level'] = pd.qcut(
        df['Daily_Usage_Time (minutes)'],
        q=4,
        labels=['Low Risk', 'Moderate Risk', 'High Risk', 'Severe Risk']
    )

    return df
```

Fig. 6: Final Data Preparation

6. Data Quality Checks

To validate the data pipeline, we confirmed that no missing values remained, data types and ranges were re-checked, and category distributions were verified. Finally, summary statistics were printed for each feature to ensure consistency and readiness for modeling.

```
def data_quality_checks(df):  
    # Check for remaining missing values  
    print("Missing values after preprocessing:")  
    print(df.isnull().sum())  
  
    # Check data types  
    print("\nFinal data types:")  
    print(df.dtypes)  
  
    # Check value ranges  
    print("\nValue ranges for numeric columns:")  
    print(df.describe())  
  
    # Check category distributions  
    print("\nCategory distributions:")  
    for col in df.select_dtypes(include=['category']).columns:  
        print(f"\n{col}:")  
        print(df[col].value_counts())
```

Fig. 7: Data Quality Checks

This preprocessing pipeline ensured that our data was clean, consistent, and ready for analysis. The steps were carefully chosen to maintain data integrity while preparing it for machine learning model training.

V. Exploratory Data Analysis

The Exploratory Data Analysis (EDA) phase was crucial for understanding the dataset's characteristics and identifying patterns in social media usage behavior. The analysis was conducted using Python's data visualization libraries and statistical methods.

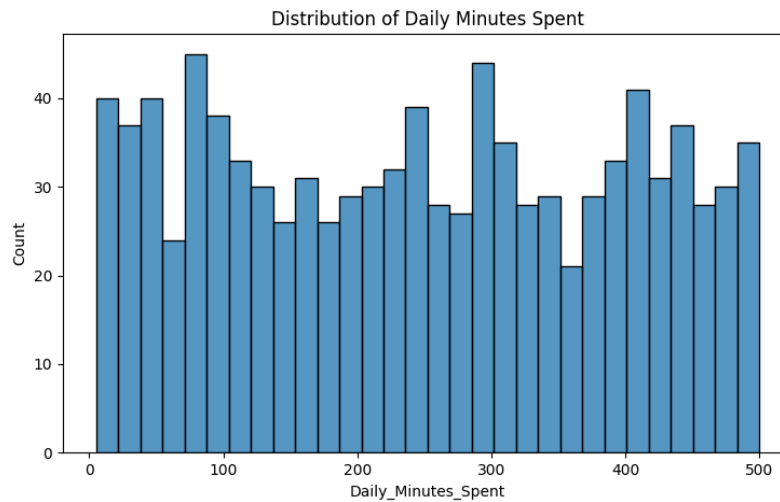
In the demographic analysis, users were categorized into five distinct age groups, and distribution analysis revealed a concentration of users within certain age ranges. These groupings helped identify specific patterns of social media usage that varied by age. Gender distribution was also analyzed to understand gender-based usage trends, comparing addiction levels across genders and revealing differences in platform preferences between male and female users.

The platform analysis section focused on usage patterns across various social media platforms. It included an examination of the time users spent on each platform and how users were distributed across them. Engagement metrics specific to each platform were also explored. In addition, addiction levels were classified for each platform to assess the relative risk of addiction, and a comparative analysis was conducted to understand platform-specific patterns of addictive behavior.

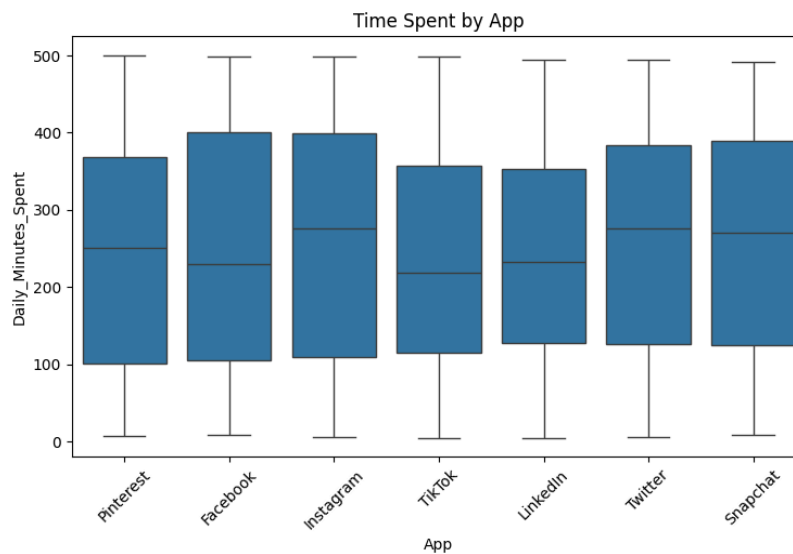
In the addiction analysis, a histogram was used to visualize the distribution of addiction scores. These scores were categorized into four levels based on percentile ranges: Low (0–25th percentile), Moderate (25–50th percentile), High (50–75th percentile), and Very High (75–100th percentile). Further analysis was conducted to study correlations between usage time and addiction levels, along with demographic influences and variations in addiction patterns by platform.

Visualizations played a key role in illustrating the findings. Distribution plots showed how addiction scores, age groups, and platform usage were spread across the dataset. Comparative insights were visualized through stacked bar charts, highlighting addiction levels by platform, platform-specific addiction patterns, and demographic breakdowns. A correlation heatmap provided more insights into relationships between different features, including platform-specific and demographic correlations.

Statistical insights added quantitative depth to the analysis, which included statistics on usage time, such as the mean daily usage, standard deviation, and range, as well as platform-specific averages. Data on user distribution and engagement across platforms was analyzed, alongside addiction level distributions. Additionally, demographic statistics were also summarized, detailing age group and gender distributions and the unique usage patterns associated with each demographic segment.

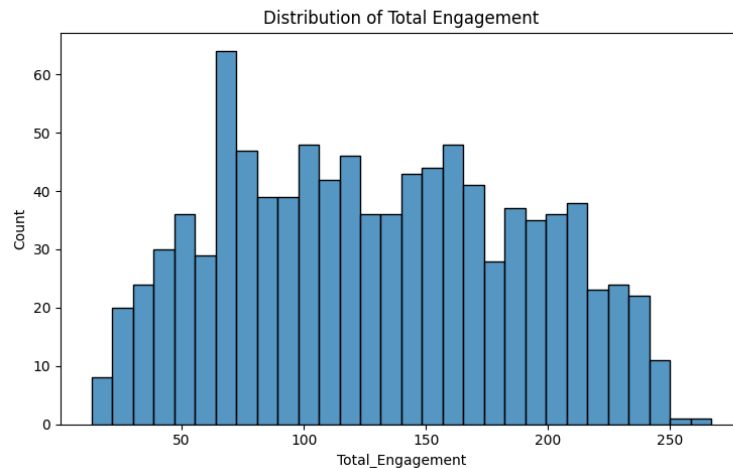


This histogram shows the distribution of daily minutes spent on social media across all users in the dataset. The x-axis represents the number of minutes spent per day, while the y-axis shows the count of users falling into each time range. The data appears fairly spread out, with no extreme skew, suggesting that users exhibit a wide variety of social media usage habits, ranging from very low to very high daily engagement.

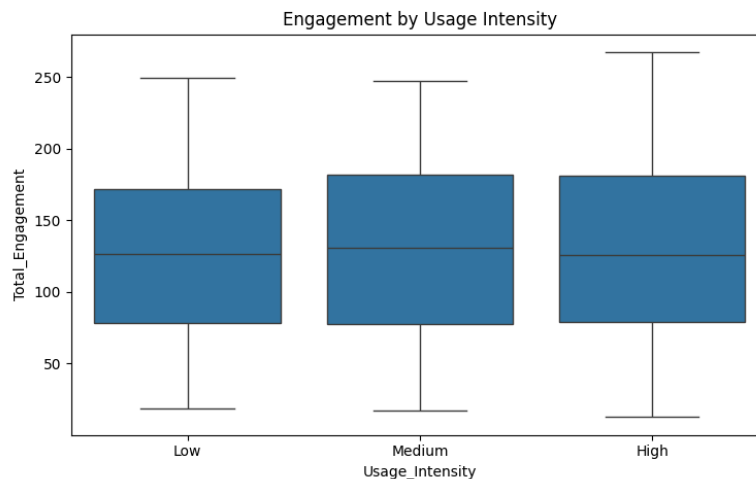


This boxplot illustrates the distribution of daily minutes spent on each social media app. Each box represents the interquartile range (middle 50%) of usage time for that platform, with the line inside showing the median. The whiskers show the range of typical usage, while any points outside are potential outliers. From the plot, it's clear that while all platforms have a wide spread

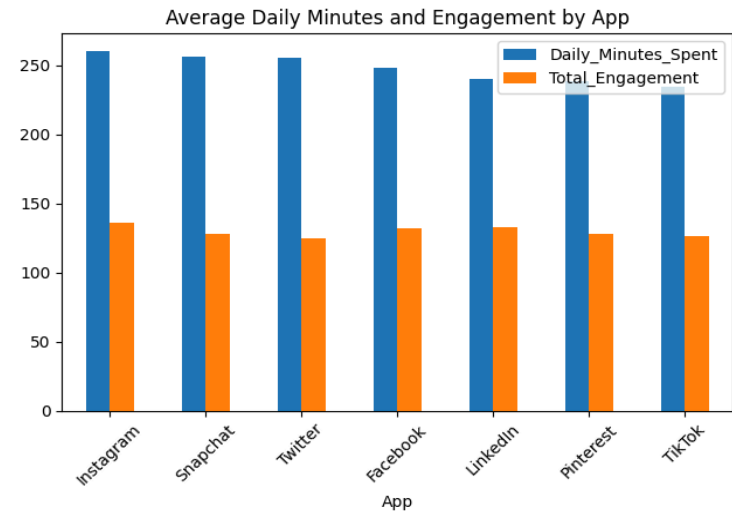
of usage times, apps like Twitter and Snapchat show slightly higher median usage compared to others, indicating users tend to spend more time on these apps daily.



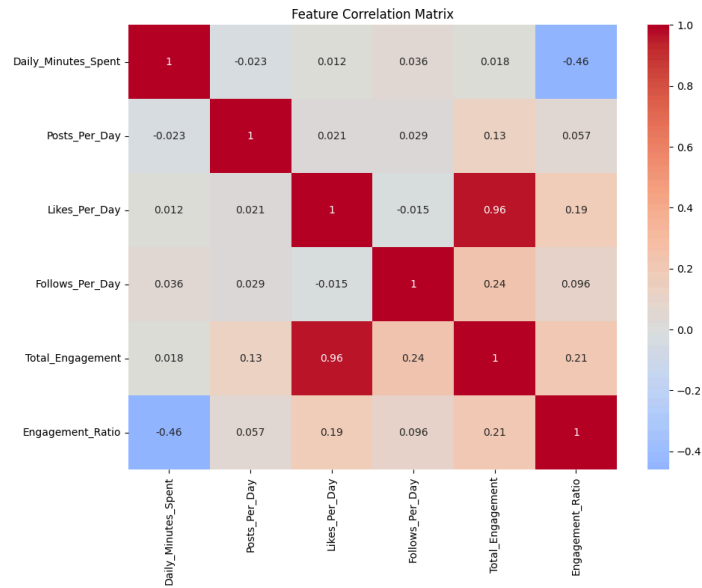
This histogram displays the distribution of total engagement across users, where engagement likely includes interactions such as likes, comments, shares, or similar activities. The x-axis represents total engagement scores, and the y-axis shows the number of users in each range. The distribution appears slightly right-skewed, indicating that while most users fall within a moderate engagement range, there are fewer users with very high total engagement. There is also a noticeable peak around the 60–70 engagement range, suggesting a common level of interaction among users.



This boxplot compares total engagement across different usage intensity levels: Low, Medium, and High. Each category shows a fairly similar spread and median, indicating that higher usage intensity does not necessarily lead to significantly higher engagement. While users in the high-intensity group have a slightly wider range of engagement values, the median engagement remains relatively consistent across all three groups, suggesting that simply spending more time on social media doesn't guarantee more interaction or activity.



This grouped bar chart compares the average daily minutes spent and total engagement across different social media apps. For each platform, the blue bar represents the average time users spend daily, while the orange bar shows the average total engagement. Instagram has the highest daily usage and engagement, followed closely by Snapchat and Twitter. Despite small variations, all platforms show a consistent trend where time spent is significantly higher than engagement, indicating that users may spend a lot of time browsing without necessarily interacting.



This chart shows how different social media behaviors are connected. There's a strong link between liking posts and having high total engagement, which means users who give more likes tend to interact more overall. Following others also slightly increases engagement. On the other hand, spending more time on social media is linked to a lower engagement ratio, suggesting that

just being online longer doesn't mean users are interacting more effectively. Most of the other features don't have strong relationships with each other.

VI. Results

The analysis revealed several key observations about social media usage and addiction. Platform preferences varied significantly across users, with certain platforms showing higher addiction levels. These preferences were often linked to age groups, and gender-specific trends were also evident. For example, female users tended to prefer certain platforms more than male users. Usage patterns showed a strong correlation between the amount of time spent on social media and addiction levels, with age and platform choice further influencing engagement behaviors.

Risk factors were identified through the data, highlighting both platform and demographic influences. The age group between 18 and 35 showed the highest addiction scores, while users over the age of 50 had the lowest. Female users were more likely to fall into higher addiction categories, whereas male users had a more balanced distribution across the levels. Among platforms, Instagram had the highest average addiction score, while LinkedIn had the lowest.

These insights suggest the importance of targeting high-risk age groups, particularly those between 18 and 35, and addressing addiction patterns specific to platforms such as Instagram and Snapchat. Gender-specific strategies and age-appropriate guidelines may also help reduce risk. Additionally, platform-specific features could be designed to discourage addictive behaviors.

Overall, the findings show that social media addiction is shaped by a combination of age, platform choice, gender, and usage patterns, especially the amount of time spent daily. Understanding these factors allows for the development of focused intervention strategies tailored to specific user groups and platforms.

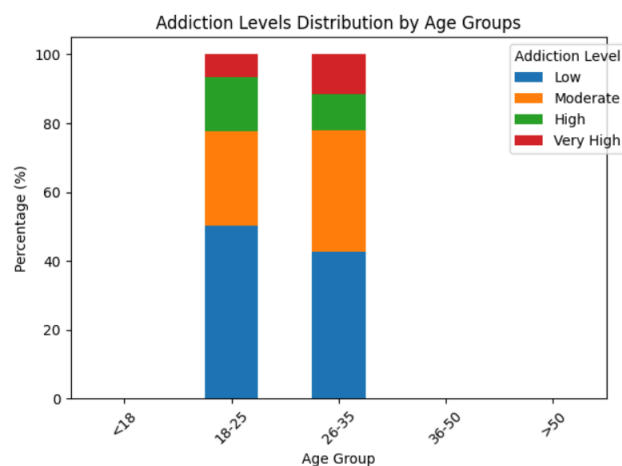


Figure (8) Addiction Levels Distribution by Age Groups

From Figure 8, this stacked bar chart shows the distribution of social media addiction levels across different age groups. The age groups 18–25 and 26–35 have a noticeably higher percentage of users in the high and very high addiction categories compared to other age groups. Both groups also have lower percentages of users in the low addiction category, highlighting them as more vulnerable to excessive social media use. There is little to no data shown for users under 18 and over 35, indicating that the analysis focused mainly on young adult populations.

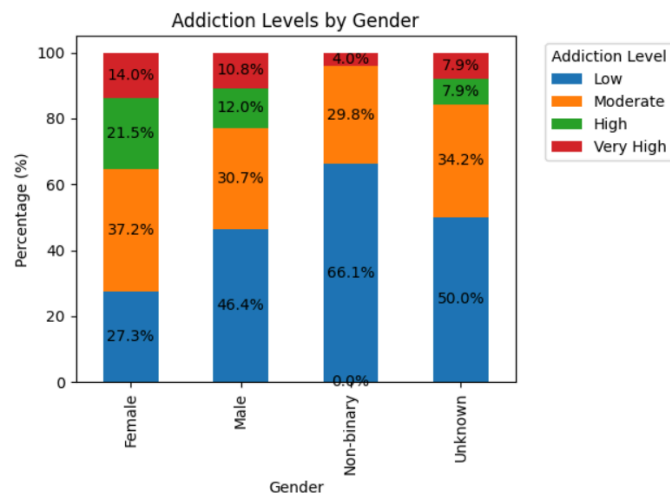


Figure (9) Addiction Levels By Gender

From Figure 9, this chart shows the distribution of addiction levels by gender. Female users have the highest combined percentage of moderate, high, and very high addiction levels, while male users show a more balanced spread, with a higher portion falling into the low category. Non-binary users mostly fall into the low addiction level, and users with unknown gender data also show a majority in the low and moderate categories.

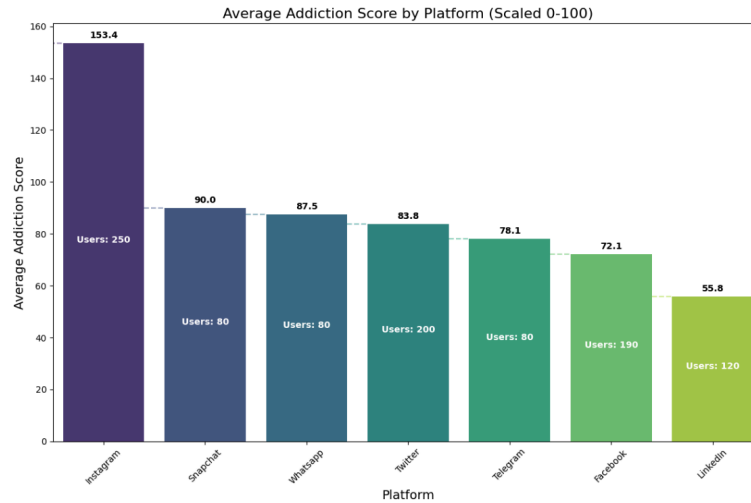


Figure (10) Average Addiction Score By Platform

From Figure 10, this bar chart shows the average addiction score for each social media platform. Instagram has the highest score by a large margin, followed by Snapchat and WhatsApp. LinkedIn has the lowest average addiction score, suggesting it is used less compulsively compared to the other platforms. These results highlight which platforms may be more likely to encourage addictive behavior.

VII. Why is our project Innovative?

In our project, we use an innovative algorithm that can be used to calculate the addiction levels. We can use it to analyze the user's data on different platforms like Facebook, Instagram, and so on. We built a relationship between the usage time and the addiction level. We also correlate the information with some factors like age and gender. This enables me to identify how social media platforms impact specific groups. Like in this project, we find that female teenagers are more likely to get into high-risk addiction.

VIII. Recommendations and further research

As social media use and misuse continue to grow worldwide, it is critical to understand how these platforms shape behavior in different populations. We recommend expanding the dataset to include users from a broader range of cultures. Doing so will help confirm whether the demographic and economic patterns we observed are consistent across regions and reveal culture-specific influences on addiction. Future studies should also examine how sudden economic changes, such as widespread job loss or recession, affect addictive usage patterns, providing evidence for targeted social support programs and economic policies. To increase predictive power, researchers can experiment with deep learning and other hybrid machine-learning ensembles, which may improve both accuracy and robustness. Finally, adding psychological measures such as loneliness, self-esteem, anxiety, and depression to these models can sharpen predictions and offer deeper insight into the emotional drivers behind social media dependence.

IX. Conclusion

This study provides a comprehensive analysis of how demographic and socioeconomic factors influence social media addiction. Through a combination of machine learning techniques and exploratory data analysis, we identified clear patterns linking age, gender, platform preferences, and income level to varying degrees of addictive behavior. Our findings reveal that younger users, particularly those in the 18-25 age group, and female users exhibit higher addiction levels, with platforms such as Instagram and Snapchat associated with the most severe risk. Conversely, older users and platforms like LinkedIn demonstrate lower addiction tendencies.

The use of models such as Logistic Regression, Random Forest, and XGBoost allowed us to accurately predict addiction risk based on behavioral and demographic features, highlighting screen time, platform engagements, and age as the most influential predictors. These insights reinforce the importance of personalized and context-sensitive interventions.

Understanding that social media addiction is not uniformly experienced across all populations emphasizes the need for platform-specific and demographic-targeted strategies. Developers, educators, mental health professionals, and policymakers can leverage these findings to design tools and programs that encourage healthier online habits, particularly among vulnerable user groups. Ultimately, our data-driven approach underscores the critical role of interdisciplinary research in promoting digital well-being in an increasingly connected world.

X. References

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